

zPSC Calibration and Test Manual

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Chapter 1

Introduction

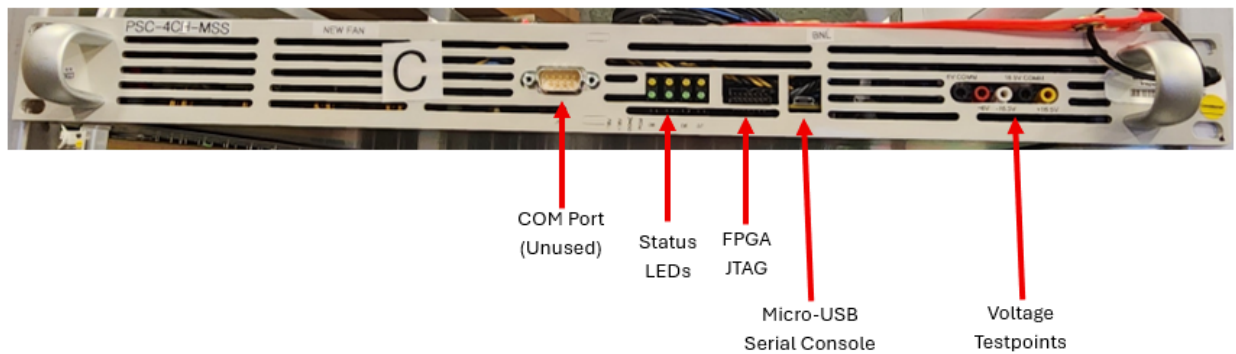
Chapter 2

zPSC Chassis Overview

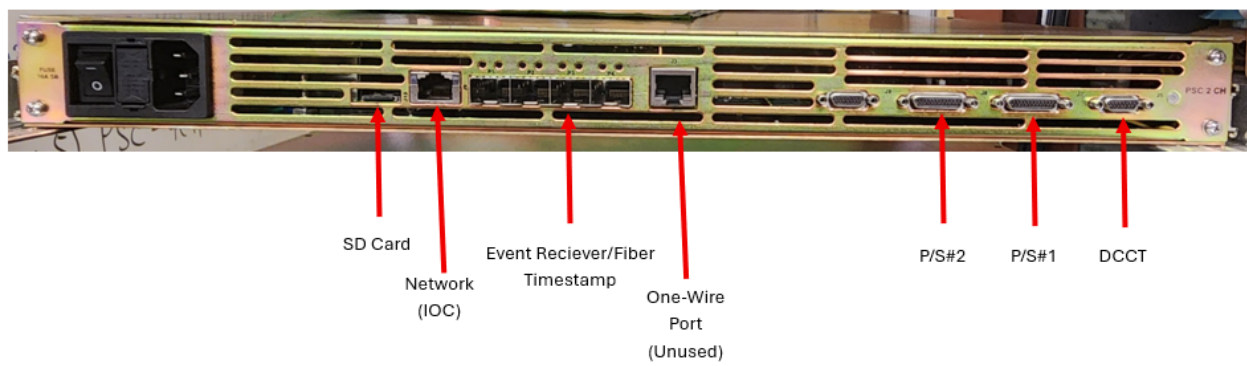
2.1 Introduction

2.2 Front Panels (2- and 4-Channel PSCs)

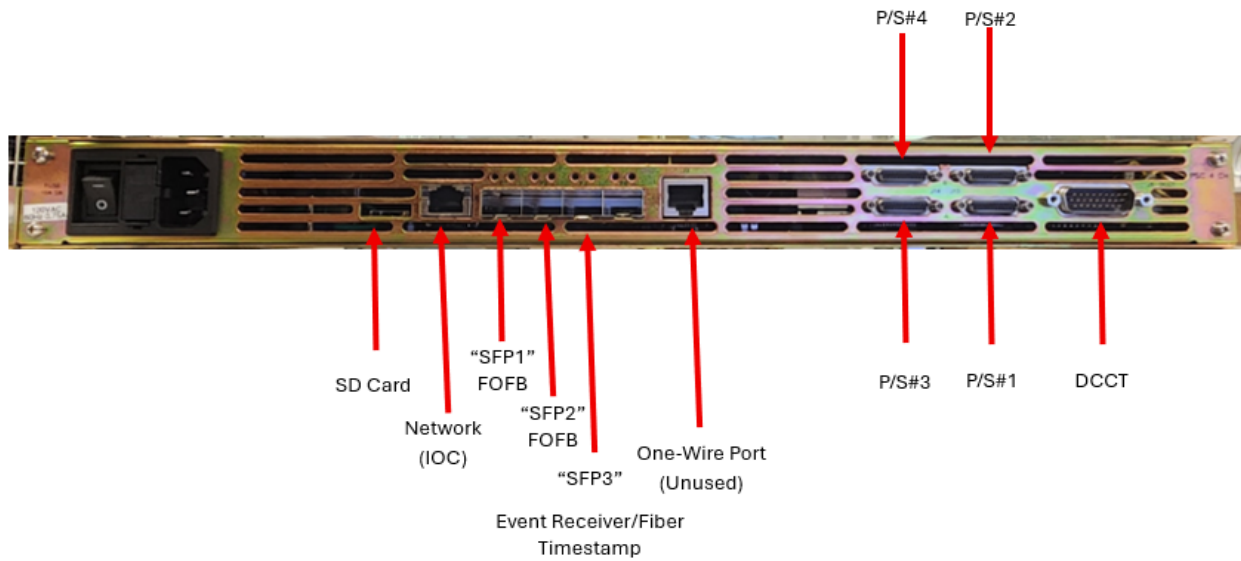
Both 2- and 4-Channel units share identical front panels:



2.3 Rear Panel - 2 Channel



2.4 Rear Panel - 4 Channel



2.5 Internal Component Layout - 2 Channel

2.6 Internal Component Layout - 4 Channel

Chapter 3

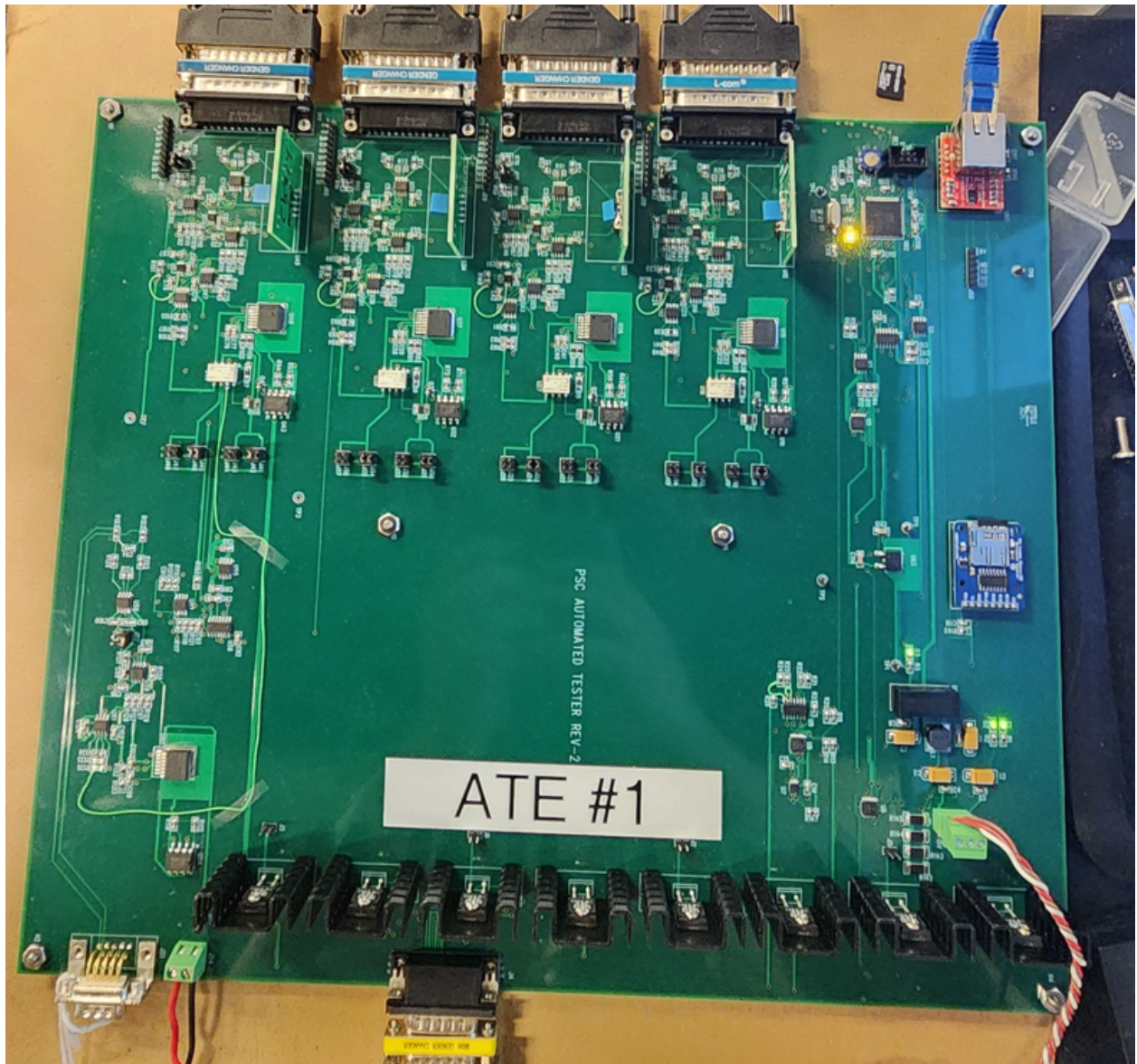
zPSC Connections

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Chapter 4

ATE Connections

4.1 Overview



On initial setup, there are a few fixed connections that need to be made to the ATE:

- Current Shunt: Connect this to a resistance standard for calibrations, or to a wire loop if not being used for current measurement
- $\pm 15V$ Supply: Connect this to a split supply, draws less than 400mA under full load
- Network: Connect to the local test network
- SD Card must be configured once, but does not require any regular changes
- Jumpers: Install jumpers in J8-9, J12-13, J19-20, JJ23-24, J30-31, J34-35, J41-42, and J45-J46 for normal operations.

For every device under test:

- Channel 1, 2, 3, and 4 Power Supply Cables: These cables connect to each of the micro-D Power Supply I/O connectors on the rear of the PSC
- Tuning Boards: Tuning boards must be installed to match the loads being tested.
- DCCT Cable must be connected for every PSC. There are two cables - one for 4-Channel units, and another for 2-Channel units.

4.2 Connections

4.2.1 At the ATE

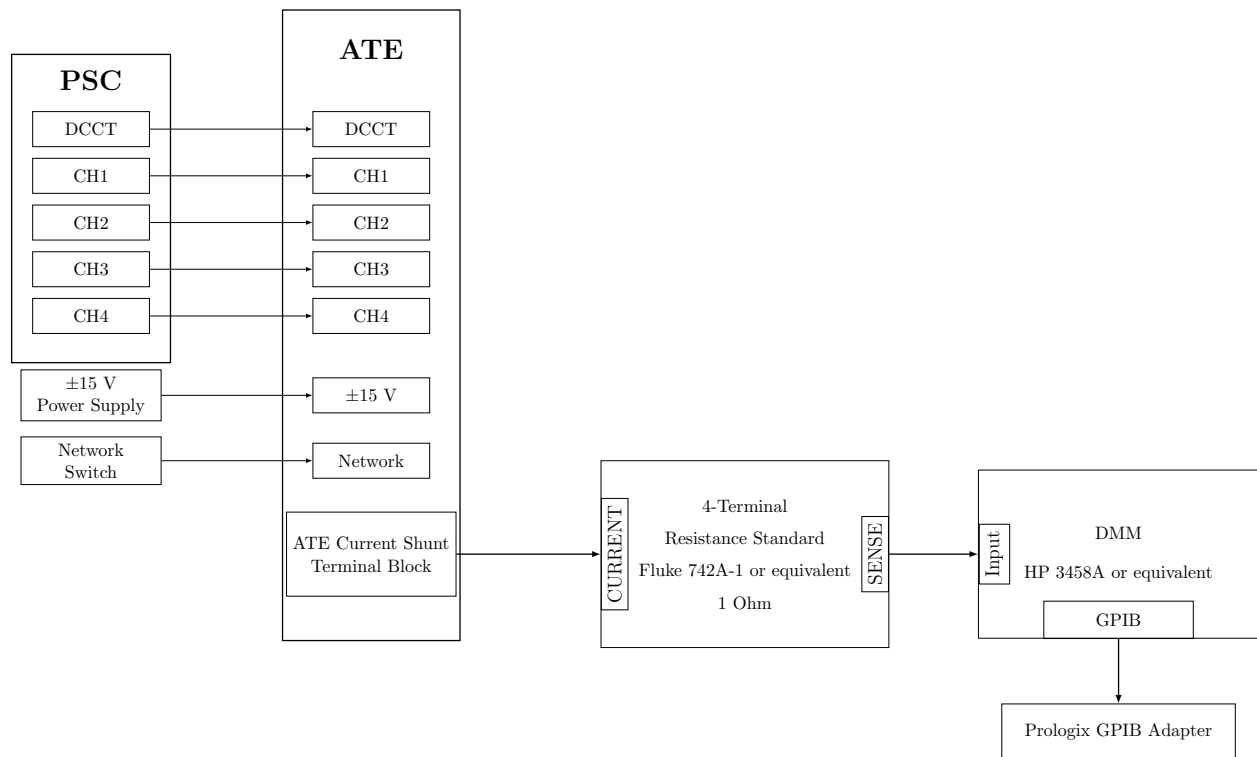


Figure 4.1: ATE Connection Block Diagram

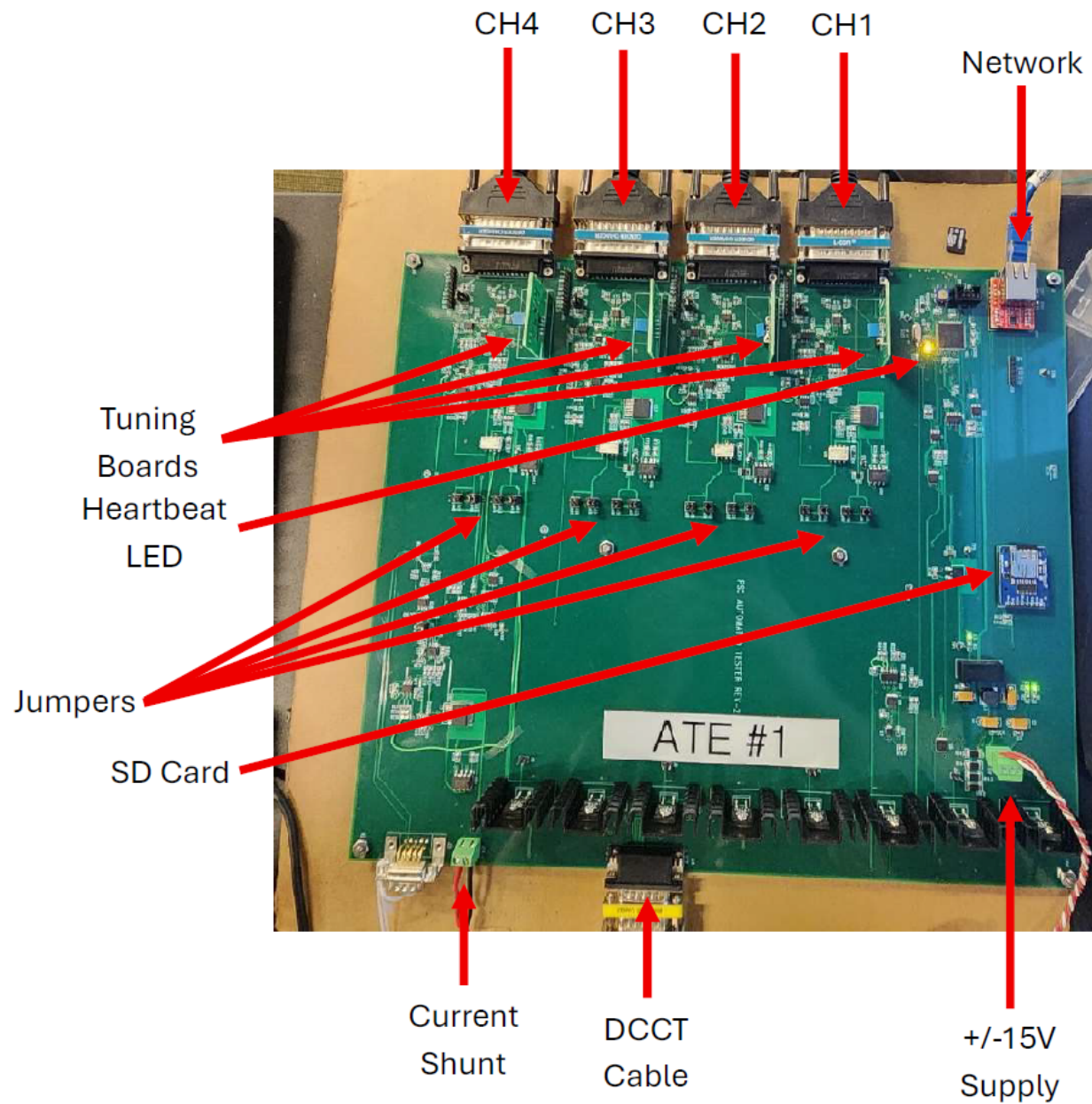


Figure 4.2: Annotated ATE Board

4.2.2 Current Measurement Loop/DMM Connection

The DMM is connected to the workstation via a GPIB adapter from Prologix. These are available in both USB and Ethernet forms.

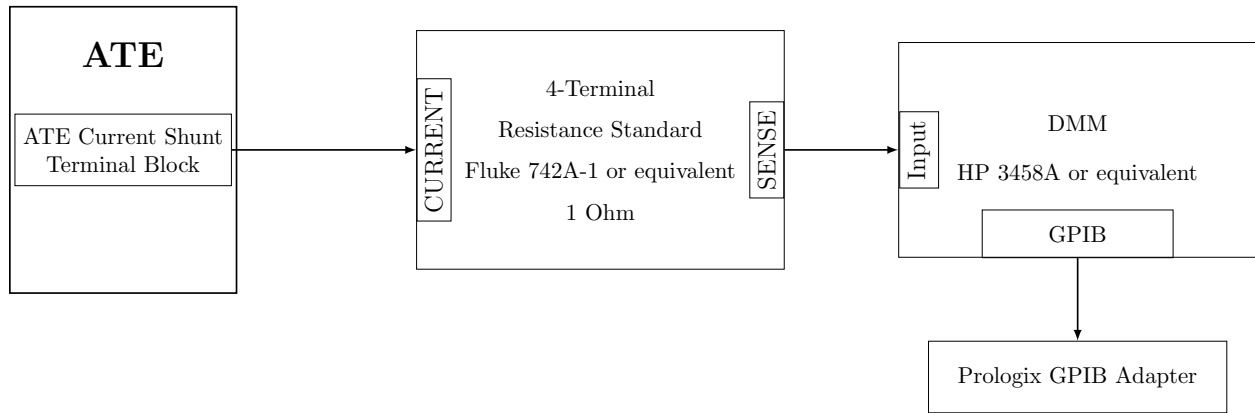


Figure 4.3: ATE → DMM Current Loop

